Per-unit and synchronous reactance

Answers and worked solutions · Day 13 · Thu 5 Nov · pack M10

5. Practice

From your course files

PQ 8.1 (practice questions).

Single-phase, so there is no $\sqrt{3}$.

a. $I_B = S_B/V_B = 2000/240 = \mathbf{8.333}$ A

b. $Z_B = V_B/I_B = 240/8.333 = \mathbf{28.80\ \Omega}$ (also $V_B^2/S_B = 57600/2000 = 28.80\ \Omega$)

c. $R_B = Z_B = \mathbf{28.80\ \Omega}$

d. $X_B = Z_B = \mathbf{28.80\ \Omega}$

Resistance, reactance and impedance all share one base. This matches the course answers.

PQ 8.2 (practice questions).

Convert to pu ($Z_B = 28.80\ \Omega$): $R = 25/28.80 = 0.8681$ pu, $X = 125/28.80 = 4.340$ pu.

(a) $\mathbf{Z} = \dfrac{1}{1/0.8681 + 1/(j4.340)} = 0.8512\angle 11.31^\circ$ pu. (In ohms this is $24.51\angle 11.31^\circ\ \Omega$, and $24.51/28.80$ gives the same value.)

(b) $\mathbf{V} = 240/240 = 1\angle 0^\circ$ pu

(c) $\mathbf{I} = \mathbf{V}/\mathbf{Z} = 1/0.8512\angle 11.31^\circ = \mathbf{1.175\angle -11.31^\circ}$ **pu**. In amps that is $1.175 \times 8.333 = 9.79$ A.

(d) pf $= \cos(11.31^\circ) = \mathbf{0.9806}$ **lag** (the current lags the voltage).

(e) $\mathbf{S} = \mathbf{V}\mathbf{I}^* = 1 \times 1.175\angle +11.31^\circ = \mathbf{1.175\angle 11.31^\circ = 1.152 + j0.2304}$ **pu**

(f) $P = 1.152 \times 2000 = \mathbf{2304}$ W. Check with method (iii): $P = V^2/R = 1^2/0.8681 = 1.152$ pu. You cannot use I^2R because I is the total current, not the resistor current.

(g) $Q = 0.2304 \times 2000 = +\mathbf{460.8}$ VAr (the course gives 460.9 from rounding).

(h) $S = 1.175 \times 2000 = \mathbf{2350}$ VA

The course gives $\mathbf{Z} = 0.8510\angle 11.31^\circ$ pu, which is $24.51/28.80$ with rounded inputs. The unrounded value is 0.8512. Both round to the same current.

PQ 8.3 (practice questions).

Bases: $I_B = 1000/200 = 5$ A. $Z_B = 200/5 = 40\ \Omega$.

Convert: $\mathbf{Z}_1 = 100/40 = 2.5\angle 0^\circ$ pu. $\mathbf{Z}_2 = (10 + j20)/40 = 0.25 + j0.5$ pu.
$\mathbf{V} = 200/200 = 1\angle 0^\circ$ pu.

SOLVE:

$$\mathbf{I}_1 = \frac{1}{2.5} = 0.4\angle 0^\circ \text{ pu}, \qquad \mathbf{I}_2 = \frac{1}{0.25 + j0.5} = 1.789\angle -63.43^\circ \text{ pu}$$

Convert back: $I_1 = 0.4 \times 5 = \mathbf{2.0}$ A, and $I_2 = 1.789 \times 5 = \mathbf{8.944}$ A.

Extra check against Q2.20(a): $\mathbf{S} = \mathbf{V}(\mathbf{I}_1 + \mathbf{I}_2)^* = 1.2 + j1.6$ pu, which is 1200 W and +1600 VAr. pf $= \cos(\tan^{-1}(1.6/1.2)) = 0.600$ lag. This matches the course answers.

PQ 8.4 (practice questions).

a. $V_{B,L} = \mathbf{11}$ kV (the chosen primary base)

b. $V_{B,ph} = 11/\sqrt{3} = \mathbf{6.351}$ kV

c. $I_B = \dfrac{10 \times 10^6}{\sqrt{3} \times 11 \times 10^3} = \mathbf{524.9}$ A

d. $Z_B = \dfrac{(11 \times 10^3)^2}{10 \times 10^6} = \mathbf{12.10\ \Omega}$. Check: $6351/524.9 = 12.10\ \Omega$.

(e), (f) $X_B = R_B = \mathbf{12.10\ \Omega}$

g. $P_B = \mathbf{10}$ MW, (h) $Q_B = \mathbf{10}$ MVAr, (i) $S_B = \mathbf{10}$ MVA

This matches the course answers.

PQ 8.5 (practice questions).

Each part has its own voltage base (the system's nominal line voltage), with the same S_B.

a. $Z_B = 3300^2/750 \times 10^3 = 14.52\ \Omega$. $X = j5/14.52 = \mathbf{j0.3444}$ pu.

b. $Z_B = 415^2/750 \times 10^3 = 0.2296\ \Omega$. $R = 0.05/0.2296 = \mathbf{0.2177}$ pu.

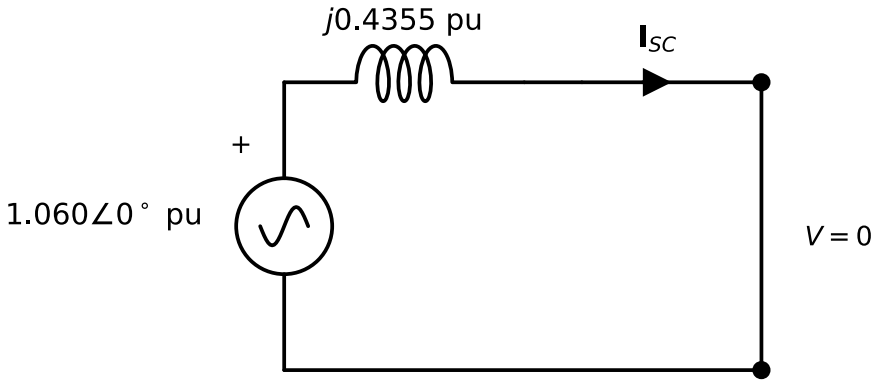
A tiny 0.05 Ω is a large 0.22 pu on a low-voltage system. That is the “perspective” the course talks about. This matches the course answers.

PQ 8.6 (practice questions).

(a) $S_B = 10$ kVA, $V_B = 415$ V (the rated voltage).

$$I_B = \frac{10 \times 10^3}{\sqrt{3} \times 415} = \mathbf{13.91}\ \text{A}, \qquad Z_B = \frac{415^2}{10 \times 10^3} = \mathbf{17.22}\ \boldsymbol{\Omega}$$

(b) $\mathbf{E} = 440/415 = \mathbf{1.060\angle 0^\circ}$ **pu**. $jX = j7.5/17.22 = \mathbf{j0.4355}$ **pu**. Draw a source of 1.060 pu in series with $j0.4355$ pu, with the output terminals shorted for part (c).



Practice Question 8.6 per-unit circuit: 1.060 pu behind j0.4355 pu with the terminals shorted

(c) $I_{SC} = 1.060/0.4355 = \mathbf{2.435}$ pu, so $I_{SC} = 2.435 \times 13.91 = \mathbf{33.87}$ A.

(d) Per phase: $E_1 = 440/\sqrt{3} = 254.0$ V. $I_{SC} = 254.0/7.5 = \mathbf{33.87}$ A, the same.

The course gives $j0.4353$ pu (from rounding Z_B up to 17.23). The unrounded value is 0.4355. The current answers agree exactly.

PQ 8.7 (practice questions).

Use $\mathbf{V} = V\angle 0^\circ$ and $\mathbf{I} = (S/V)\angle \pm \cos^{-1}(\text{pf})$. Use $-$ for lagging and $+$ for leading.

(a) $\mathbf{V} = 1\angle 0^\circ$ pu. $\cos^{-1} 0.9 = 25.84^\circ$. $\mathbf{I} = (1/1)\angle -25.84^\circ = \mathbf{1\angle -25.84^\circ}$ **pu**.

(b) $\mathbf{V} = 1.05\angle 0^\circ$ pu. $\mathbf{I} = (0.5/1.05)\angle +25.84^\circ = \mathbf{0.4762\angle +25.84^\circ}$ **pu**.

(c) $P = 100/150 = 0.6667$ pu. $S = 0.6667/0.85 = 0.7843$ pu. $\mathbf{V} = 1\angle 0^\circ$ pu. $\cos^{-1} 0.85 = 31.79^\circ$. $\mathbf{I} = \mathbf{0.7843\angle -31.79^\circ}$ **pu**. (The course gives 0.7844, a rounding difference.)

(d) $\mathbf{S} = (150 + j50)/200 = 0.75 + j0.25$ pu, so $|\mathbf{S}| = 0.7906$ pu. $\mathbf{V} = 0.95\angle 0^\circ$ pu.

$$\mathbf{I} = \left(\frac{\mathbf{S}}{\mathbf{V}}\right)^* = \frac{0.7906}{0.95}\angle -\tan^{-1}(0.25/0.75) = \mathbf{0.8322\angle -18.43^\circ}\ \text{pu}$$

The course gives −18.44°. The exact value of $\tan^{-1}(1/3)$ is 18.435°, so both roundings are fine.

Jun23 Q4a (past exam, 5 marks).

(i) Same I_F = 60 A for both readings.

$$E_{0,ph} = \frac{4600}{\sqrt{3}} = 2656\ \text{V}, \qquad |Z_S| = \frac{2656}{720} = 3.689\ \Omega$$

$$X_S = \sqrt{3.689^2 - 0.3^2} = \mathbf{3.676\ \Omega}$$

Stop here. The question asks for ohms. (For reference, $Z_B = 14^2/150 = 1.307\ \Omega$, so $X_S = 2.814$ pu.)

(ii) Rated voltage, so $\mathbf{V} = \mathbf{1.0\angle 0^\circ}$ **pu**.

$$\mathbf{S} = \frac{100 + j50}{150} = 0.6667 + j0.3333\ \text{pu}, \quad |\mathbf{S}| = 0.7454\ \text{pu}$$

The generator delivers Q > 0, so its pf is lagging and the current lags:

$$\mathbf{I} = \left(\frac{\mathbf{S}}{\mathbf{V}}\right)^* = 0.7454\angle -\tan^{-1}(0.5) = \mathbf{0.7454\angle -26.57^\circ}\ \text{pu}$$

(In amps: $I_B = 150 \times 10^6/(\sqrt{3} \times 14 \times 10^3) = 6186$ A, so $I = 4611$ A.)

Course answer: $X_S = 3.676\ \Omega$; $\mathbf{V} = 1.0\angle 0^\circ$ pu, $\mathbf{I} = 0.7453\angle -26.57^\circ$ pu. The text copy of the answer file loses the minus sign, but the PDF shows −26.57°. $\sqrt{0.6667^2 + 0.3333^2} = 0.74536$, so 0.7454 is the correctly rounded value; the course truncates to 0.7453.

Jun24 Q4a (past exam, 5 marks).

(i) The OC voltage is a line value (the course convention):

$$|Z_S| = \frac{12000/\sqrt{3}}{6500} = 1.066\ \Omega, \qquad X_S = \sqrt{1.066^2 - 0.2^2} = 1.047\ \Omega$$

$$Z_B = \frac{(14 \times 10^3)^2}{400 \times 10^6} = 0.4900\ \Omega, \qquad X_S = \frac{1.047}{0.4900} = \mathbf{2.137}\ \text{pu}$$

(Also $R = 0.2/0.49 = 0.408$ pu.) The marks are for: $\sqrt{3}$ and $|Z_S|$ (1), removing R (1), and Z_B with the pu conversion (1).

(ii) $\mathbf{V} = \mathbf{0.97}\angle\mathbf{0}^\circ$ pu.

$P = 200/400 = 0.5$ pu. $S = 0.5/0.8 = 0.625$ pu. $I = 0.625/0.97 = 0.6443$ pu. The pf is lagging, so the angle is $-\cos^{-1} 0.8 = -36.87^\circ$.

$$\mathbf{I} = \mathbf{0.6443}\angle\mathbf{-36.87}^\circ \text{ pu}$$

Course answer: $X_S = 2.137$ pu ($X_S = 1.047\ \Omega$); $\mathbf{V} = 0.97\angle 0^\circ$ pu, $\mathbf{I} = 0.6443\angle -36.87^\circ$ pu. This matches exactly.

PQ 8.11 (practice questions; parts (a)–(b) here).

a. $Z_B = \frac{15^2}{250} = 0.9\ \Omega$. $|Z_S| = \frac{5750/\sqrt{3}}{2050} = 1.619\ \Omega$.
$X_S = \sqrt{1.619^2 - 0.15^2} = 1.612\ \Omega$, so $X_S = 1.791$ **pu**.

b. Full load means S = 1 pu at V = 1 pu, so $|I| = 1$ pu. 0.85 lag: $\mathbf{I} = 1\angle -31.79^\circ$ **pu**.

Both match the course answers (1.791 pu, $1\angle -31.79^\circ$ pu). Parts (c)–(f) are worked in M11.

PQ 8.12 (practice questions; parts (a)–(b) here).

a. $Z_B = \frac{16^2}{120} = 2.133\ \Omega$. $|Z_S| = \frac{4400/\sqrt{3}}{600} = 4.234\ \Omega$.
$X_S = \sqrt{4.234^2 - 0.1^2} = 4.233\ \Omega$, so $X_S = 1.984$ **pu**.

b. $P = 102/120 = 0.85$ pu. $S = 0.85/0.95 = 0.8947$ pu. At V = 1 pu:
$\mathbf{I} = 0.8947\angle -18.19^\circ$ **pu**.

Course answers: 1.984 pu, $0.8947\angle -18.2^\circ$ pu. Part (c) is worked in M11.

New questions

New Question 1 (easy) [4 marks].

a. $I_B = \dfrac{50 \times 10^6}{\sqrt{3} \times 11 \times 10^3} = \mathbf{2624}$ A. $Z_B = \dfrac{11^2}{50} = \mathbf{2.420\ \Omega}$.

b. $X = 0.8/2.420 = \mathbf{0.3306}$ pu.

c. Line: $0.95 \times 11 = \mathbf{10.45}$ kV. Phase: $10.45/\sqrt{3} = \mathbf{6.033}$ kV. The pu value is the same for both; only the base differs.

New Question 2 (easy–medium) [5 marks].

a. Take $I_F = 1$ A: $V_{OC} = 400$ V line, so $E_{0,ph} = 230.9$ V, and $I_{SC} = 80$ A.

$$|Z_S| = 230.9/80 = 2.887\ \Omega, \qquad X_S = \sqrt{2.887^2 - 0.05^2} = \mathbf{2.886\ \Omega}$$

$Z_B = 18^2/300 = 1.080\ \Omega$, so $X_S = 2.886/1.080 = \mathbf{2.673}$ pu.

b. $I_F = 18000/400 = \mathbf{45}$ A (this is the field current for $E_0 = 1$ pu).

c. $I_F = 1.8 \times 45 = \mathbf{81}$ A.

New Question 3 (medium) [4 marks].

$\mathbf{V} = 1.03\angle 0^\circ$ pu.

A generator that absorbs Q has $Q < 0$: $\mathbf{S} = (180 - j60)/250 = 0.72 - j0.24$ pu.
$|\mathbf{S}| = 0.7589$ pu.

$$\mathbf{I} = \left(\frac{\mathbf{S}}{\mathbf{V}}\right)^* = \frac{0.7589}{1.03}\angle +\tan^{-1}(0.24/0.72) = \mathbf{0.7368\angle +18.43^\circ}\ \text{pu}$$

pf $= \cos 18.43^\circ = \mathbf{0.9487}$ **leading** (the current leads the voltage, as for a generator absorbing VArs).

New Question 4 (exam-style, chained) [9 marks].

(a) Bases: $I_B = 180 \times 10^6/(\sqrt{3} \times 15 \times 10^3) = 6928$ A. $Z_B = 15^2/180 = 1.250\ \Omega$.

$$|Z_S| = \frac{6000/\sqrt{3}}{2400} = 1.443\ \Omega, \qquad X_S = \sqrt{1.443^2 - 0.12^2} = 1.438\ \Omega$$

$$X_S = 1.438/1.250 = \mathbf{1.151}\ \text{pu}, \qquad R = 0.12/1.250 = \mathbf{0.096}\ \text{pu}$$

(b) $\mathbf{V} = \mathbf{1.02\angle 0^\circ}$ **pu**. $P = 135/180 = 0.75$ pu, so $S = 0.75/0.9 = 0.8333$ pu.

$$\mathbf{I} = \frac{0.8333}{1.02}\angle -25.84^\circ = \mathbf{0.8170\angle -25.84^\circ}\ \text{pu}, \qquad I = 0.8170 \times 6928 = \mathbf{5660}\ \text{A}$$

(c) Generator convention:

New Question 4 circuit in pu: E0 behind 0.096 pu and j1.151 pu, V = 1.02 pu, I = 0.8170 at -25.84 degrees out

$$\mathbf{E}_0 = 1.02 + 0.8170\angle -25.84^\circ \times (0.096 + j1.151) = \mathbf{1.706\angle 28.42^\circ}\ \text{pu}$$

Line value: $1.706 \times 15 = \mathbf{25.59}$ kV.

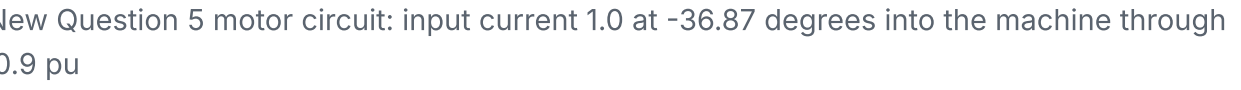
Excitation scale: $I_F(E_0 = 1 \text{ pu}) = 80 \times 15/6.0 = 200$ A. So $I_F = 1.706 \times 200 = \mathbf{341}$ A.

(If you neglect R, you get $\mathbf{E}_0 = 1.661\angle 30.62^\circ$ pu. Only neglect R when the question says so.)

New Question 5 (motor convention) [5 marks].

(a) $I_B = 500 \times 10^3/(\sqrt{3} \times 415) = 695.6$ A. $P = 400/500 = 0.8$ pu, so $S = 0.8/0.8 = 1.0$ pu. The input current lags:

$$\mathbf{I}_m = \frac{1.0}{1.0}\angle -36.87^\circ = \mathbf{1.0\angle -36.87^\circ} \text{ pu} = 0.8 - j0.6 \text{ pu}, \qquad I_m = \mathbf{695.6} \text{ A}$$



New Question 5 motor circuit: input current 1.0 at -36.87 degrees into the machine through j0.9 pu

(b) $\mathbf{E}_0 = \mathbf{V} - jX_S\mathbf{I}_m = 1 - j0.9(0.8 - j0.6) = 1 - 0.54 - j0.72 = 0.46 - j0.72$

$$\mathbf{E}_0 = \mathbf{0.8544\angle -57.43^\circ} \text{ pu}, \qquad E_0 = 0.8544 \times 415 = \mathbf{354.6} \text{ V (line)}$$

(c) $\delta = -57.43^\circ < 0$, so $\mathbf{E}_0$ lags $\mathbf{V}$, which confirms motoring. Also $E_0 < V$, so the machine is under-excited, which goes with a lagging input pf (see M11).

6. Quick recall check

Question

1. $Z_B = 16^2/120 = 2.133\ \Omega$.
2. $|Z_S| = (4400/\sqrt{3})/600 = 4.234\ \Omega$ (this is Q8.12's value).
3. $S = 0.6/0.8 = 0.75$ pu, so $\mathbf{I} = 0.75\angle +36.87^\circ$ pu.
4. Generator: $\mathbf{E}_0 = \mathbf{V} + \mathbf{I}(R + jX_S)$. Motor: $\mathbf{E}_0 = \mathbf{V} - \mathbf{I}_m(R + jX_S)$, with $\delta < 0$.
5. In the SC test the stator field is nearly equal and opposite to the rotor field, so the net flux is small and the iron does not saturate. In the OC test there is no stator current, so the full rotor field is in the iron.